

DAY - 14

SEAT NUMBER

2023 VIII 04

1100

V - 164

(E)

COMPUTER SCIENCE
PAPER - I (D-9)

Time : 3 Hours

4 Pages

Max. Marks : 50

- Instructions :**
- (1) All questions are compulsory.
 - (2) Figures to the right indicate full marks.
 - (3) Use of any type of calculator not allowed.
 - (4) Draw neat diagram wherever necessary.

1. (A) Select the correct alternative and rewrite the following :
 - (a) ALIGN is not an attribute used with _____ tag. 1
 - (i) <TR>
 - (ii) <HR>
 - (iii) <TABLE>
 - (iv) <BODY>
 - (b) int data type in c++ takes _____ bytes in memory. 1
 - (i) 2
 - (ii) 4
 - (iii) 1
 - (iv) 8
 - (c) The maximum number of comparisons for binary search to locate an ITEM from DATA array which includes 200 elements will be _____. 1
 - (i) 8
 - (ii) 7
 - (iii) 5
 - (iv) 20

- (d) A special subroutine written for I/O device such a subroutine is called _____ 1
- (i) device driver
 - (ii) application software
 - (iii) language processors
 - (iv) system software
- (B) Answer **any two** of the following :
- (a) What is File System ? List and explain the types of office system used in operating system. 3
 - (b) Write an algorithm to insert an element into a linear array. 3
 - (c) What is in-line function ? Explain with a suitable example. 3
2. (A) Answer **any two** of the following :
- (a) Explain the following HTML tags with an example of each : 3
 - (i) <HR>
 - (ii)
 - (iii) <SUP>
 - (b) What is a Computer Virus ? State various types of viruses and the basis on which they are classified. 3
 - (c) What is a Linked List ? Explain it with a suitable example. 3
- (B) Answer **any one** of the following :
- (a) Define the terms - constructor and Destructor. Write any two different characteristics of constructor and destruction. 4
 - (b) Explain the following concepts related to virtual memory : 4
 - (i) Locality of reference
 - (ii) Page fault
 - (iii) Working set
 - (iv) Page replacement policy

3. (A) Answer **any two** of the following :
- (a) What is a class in C++ ? Explain it with a suitable example. 3
 - (b) Define the following terms with respect to binary tree : 3
 - (i) Root
 - (ii) Child
 - (iii) Path
 - (c) With reference to Process Management, explain the following terms : 3
 - (i) Internal Priority
 - (ii) Time slice
 - (iii) External Priority
- (B) Answer **any one** of the following :
- (a) What is an Operator Function ? Give and explain the syntax of operator function. 4
 - (b) Give any four system calls of Process Management and Information Management each. 4
4. (A) Answer **any two** of the following :
- (a) Enlist the basic data types used in C++ with size of data in terms of bytes for each. 3
 - (b) Draw the tree diagram (binary tree) for following expression $(M+P)^2/(Q-d)^3$. 3
 - (c) Explain use of scope resolution operator with suitable example. 3
- (B) Answer **any one** of the following :
- (a) Explain any four various types of attacks on security system. 4
 - (b) Write an algorithm for bubble sort in linear array with N elements. 4
5. Answer **any two** of the following :
- (a) Write a C++ program that will read line of text and count the total Number of words in text. 5
 - (b) Write C++ program to find sum of the contents of an array of length 10 elements using pointers. 5

- (c) Write HTML code for the following output :

Department of Education :		
Primary	Secondary	High-Sec.

Contact Website of Education

Primary is clicking link available by web file "Primary.html". Secondary is a clicking link available by web file "Secondary.html" and High-sec is clicking link available by web file "Higher.html" on clicking "Website of Education" we can invoke a website. "www.mahaeducaton.gov.in".

OR

5. Attempt **any two** of following :

- (a) Write C++ program to find largest of four given integers using max()
int max (int, int, int, int); 5
- (b) Write C++ program to initialize the array of 10 integers and find the sum of all elements of the array. 5
- (c) Write HTML code for the following Output : 5

● Faculties of computer

- (i) Computer Science
- (ii) Computer Application
- (iii) Computer Engineering

○ Faculties of Biology

- (1) Biotechnology
- (2) Agriculture
- (3) Fisheries

□ Faculties of Maths

- (a) Pure mathematics
- (b) Applied mathematics